

Rescaling cohesive element properties for mesh independent fracture simulations

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ABSTRACT

In finite element simulations of fracture, cohesive elements introduce artificial compliance into the model leading to changes in properties such as the effective stiffness, effective speed of sound, and crack propagation speed. In this paper, the relation between effective material properties and cohesive element properties is determined analytically. Based on this, a simple method for rescaling the cohesive element thickness is proposed to reduce the effect of artificial compliance and to obtain mesh independent results. The effectiveness of this approach is demonstrated with four example problems: uniaxial stretching, beam bending, bar impact, and Kalthoff plate impact.

1. Introduction

Linear elastic fracture mechanics (LEFM) predicts a singular stress field at the tip of a crack. In reality material nonlinearity and large strains limit the magnitude of the stress. To model this behavior, Dugdale [1] and Barenblatt [2] independently introduced the notion of a *cohesive zone* (CZ) ahead of the crack tip where a different constitutive relation is introduced to eliminate the singularity. The response of the CZ is described by a *cohesive law* that relates the stress acting on the zone to the separation and/or shear across it. A simple cohesive law that allows for fracture is a bilinear relation in which the stress grows linearly from zero to a maximum stress (tensile strength $\sigma_{\max}$ in mode-I fracture, shear strength $\tau_{\max}$ in mode-II and III fracture), and then decays linearly to zero at which point the material is fractured. Other cohesive laws with different functional forms have been proposed including an exponential law [3,4], a rigid-linear law [5], and mixed-mode laws in which the fracture modes are coupled. A detailed review of cohesive laws is given in [6].

CZs have been incorporated into the finite element method (FEM) to approximate the fracture process. Special *cohesive elements* are inserted between bulk elements in places where fracture is allowed. An early example was the work of Hillerborg et al. [7] who used this approach to simulate the fracture of quasi-brittle materials such as concrete. More recent examples include simulations of composite delamination [8,9], and adhesive joint failure [3,10]. In these applications fracture is confined to specific locations where the cohesive elements are placed (such as between a fiber and a matrix in a composite). To simulate arbitrary crack propagation in bulk systems, Xu and Needleman [4] extended the CZ-FEM approach by placing cohesive elements between all bulk elements. Thus any two elements can separate to form or extend a crack when the cohesive element between them fails. Alternatively, Camacho and Ortiz [5] proposed a method where new cohesive elements are introduced as needed by duplicating nodes along previously coherent element boundaries to simulate impact damage in brittle materials. In these methods, although crack propagation is limited to lie along element edges in two dimensions (or faces in three dimensions), arbitrary cracking can be simulated if the mesh is sufficiently refined. Cohesive elements can also be used to study other mechanics applications involving interfaces, such as adhesives, interfacial

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debonding, and gaskets [11].

In addition to the cohesive element approach, other methods have been proposed to simulate crack propagation in materials. In the extended finite element method (XFEM) of Belytschko and co-workers [12,13] cracks are represented independently of the mesh, and the FEM formulation is enriched based on the LEFM crack-tip solution. In principle, XFEM allows for cracks of any shape and is efficient since mesh refinement is not required near the crack tip. The method is well-suited for fracture in pure materials, but becomes more complex for heterogeneous materials (such as interfacial fracture in bimetals). Phase field methods for modeling fracture have been developed over the past decade, see for example [14–16]. A scalar phase-field function is associated with the material, which is set to zero in broken areas and unity away from cracks, and is evolved based on a thermodynamically consistent variational framework.

Although both XFEM and phase field methods are excellent choices for simulating fracture, CZ-FEM continues to be attractive due to its simplicity, the clear physical meaning of the cohesive law, and its widespread availability in commercial FEM packages including ANSYS, LS-DYNA and Abaqus. However, a serious downside of CZ-FEM is that the addition of cohesive elements introduces an artificial compliance into the model that leads to mesh dependence in the results, especially when cohesive elements are inserted between all bulk elements. This changes both static and dynamic properties of the model with mesh resolution. The additional compliance decreases the effective Young's modulus E_{eff} of the model, and changes the effective speed of sound c_{eff} and the speed of crack propagation. The issue of mesh dependence in CZ-FEM was studied by Klein et al. [17] for a one-dimensional (1D) model where it was shown that artificial compliance was proportional to the element size h and inversely proportional to the cohesive element modulus K . This study implied that artificial compliance was inevitable and became bigger as the mesh was refined. Falk et al. [18] showed that cohesive element lengths can influence crack branching behavior. A recent study by Tabiei et al. [19] investigated a “CZ enlargement” approach for mesh independent simulations in which the CZ length is increased by scaling down the cohesive strength (σ_{max}) in the vicinity of the crack tip.

In this paper, we show that by appropriately scaling the cohesive element thickness with mesh size it is possible to maintain constant values of E_{eff} and c_{eff} with mesh refinement and obtain mesh independent results. This approach is consistent with the banded crack method of Bažant et al. [20] in which the dissipated energy due to crack formation is kept constant. The problem is first discussed theoretically in a 1D setting and then studied numerically for four problems: (1) uniaxial stretching of uniform and nonuniform meshes; (2) beam bending of uniform meshes; (3) uniaxial dynamic loading of uniform meshes; and (4) a Kalthoff plate impact problem. The simulations are performed using the Abaqus FEM code version 6.12 [11].

2. 1D analysis of cohesive element effects on FEM model properties

2.1. Cohesive laws

Different FEM codes implement cohesive elements in different ways. We focus on Abaqus in which the mode-I fracture bilinear cohesive law (see Fig. 1(a)) is defined as

$$\sigma = \begin{cases} K \epsilon & \epsilon < \epsilon_{\text{peak}}, \\ \sigma_{\text{max}} - K_e (\epsilon - \epsilon_{\text{peak}}) & \epsilon \geq \epsilon_{\text{peak}}, \end{cases} \quad (1)$$

where σ and ϵ are the stress and strain normal to the cohesive element, K and K_e are the cohesive moduli during damage initiation and damage evolution, σ_{max} and ϵ_{peak} are the stress and strain at the maximum loading the element can sustain. The area under the curve is the mode-I fracture energy G_{Ic} divided by the nominal cohesive element thickness d . The element properties are defined in Abaqus by specifying K , σ_{max} , G_{Ic} or equivalently K , σ_{max} , $u_{\text{max}} = \epsilon_{\text{max}} d$, where ϵ_{max} is the strain at fracture and u_{max} is the maximum elongation of the cohesive element in the normal direction at fracture. A corresponding cohesive law for shearing is also defined with notation shown in Fig. 1(b).

It is important to note that in Abaqus the nominal thickness d associated with a cohesive element is not the geometric thickness of

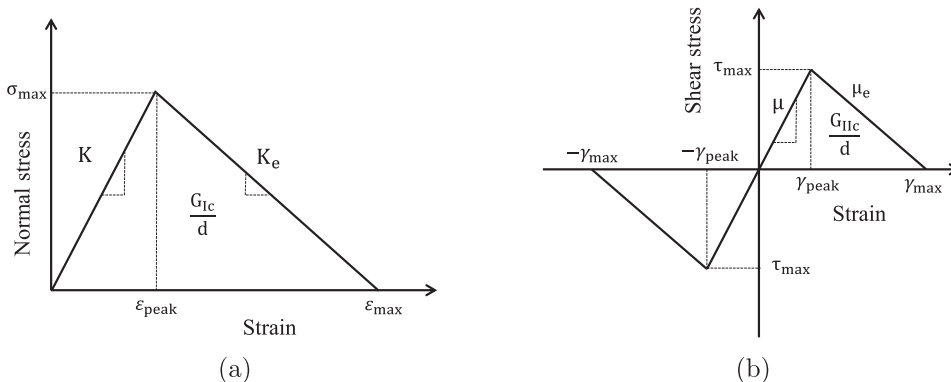


Fig. 1. Bilinear cohesive laws in (a) tension, and (b) shear.

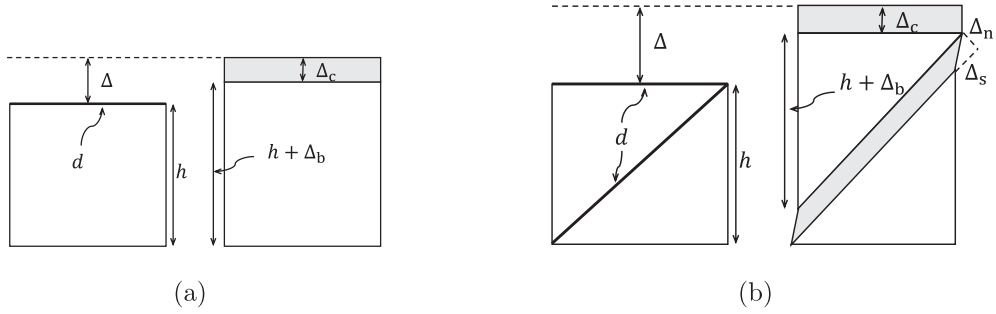


Fig. 2. Two types of 1D periodic cells. (a) One square bulk element and one cohesive element before (left) and after (right) uniaxial stretching. (b) Two triangular bulk elements separated by a cohesive elements at 45° and 0° before (left) and after (right) uniaxial stretching.

the element, but an artificial reference thickness relating elongation to strain. The geometric thickness is taken to be zero in the reference configuration.

2.2. Static properties in 1D: effective stiffness

To explore the effect of adding cohesive elements to an FEM model, we consider a 1D periodic cell containing one square bulk element with initial length h and one cohesive element with artificial thickness d as depicted in Fig. 2(a). Since the cell is periodic, this represents an infinite repeating sequence of bulk elements separated by cohesive elements. The periodic cell is stretched uniformly by an amount Δ , which is accommodated by extensions in the bulk and cohesive elements:

$$\Delta = \Delta_b + \Delta_c. \quad (2)$$

At equilibrium the stress in the bulk element and cohesive element are equal. Taking this value to be σ , and assuming a linear elastic response for the bulk element, we have

$$\sigma = E \frac{\Delta_b}{h} = K \frac{\Delta_c}{d}, \quad (3)$$

where E is Young's modulus in the bulk material. Solving for Δ_b and Δ_c in Eq. (3) and substituting into Eq. (2) gives

$$\Delta = \sigma \left(\frac{h}{E} + \frac{d}{K} \right). \quad (4)$$

The strain relative to the reference configuration is $\epsilon = \Delta/h$. Therefore the effective Young's modulus E_{eff} of the periodic element is

$$E_{\text{eff}} \equiv \frac{\sigma}{\epsilon} = \frac{\sigma h}{\Delta} = \frac{EK h}{Kh + Ed} = E \frac{1}{1 + \frac{E d}{K h}}, \quad (5)$$

which gives

$$\frac{E_{\text{eff}}}{E} = \frac{1}{1 + \frac{E d}{K h}}. \quad (6)$$

This relation shows that the introduction of cohesive element necessarily reduces the stiffness of the model. The effect of cohesive element compliance is eliminated in the limit $d \rightarrow 0$ or $K \rightarrow \infty$.

A similar derivation can be performed for the element configuration shown in Fig. 2(b) where an inclined cohesive element at 45 degrees is also present (see [supplementary material for details](#)). In this case, the E_{eff}/E ratio is

$$\frac{E_{\text{eff}}}{E} = \frac{1}{1 + \frac{(4 + \sqrt{2}) E d}{4 K h} + \frac{\sqrt{2} E d}{4 \mu h}}, \quad (7)$$

where μ is the shear modulus across the cohesive element. For the special case where $\mu = K/2$ (corresponding to a Poisson's ratio of zero) used in the numerical examples below, the result is

$$\frac{E_{\text{eff}}}{E} = \frac{1}{1 + \frac{4 + 3\sqrt{2} E d}{4 K h}}. \quad (8)$$

2.3. Dynamic properties in 1D: effective speed of sound

Next we consider the effect of cohesive elements on the dynamics of a 1D periodic system. We take both bulk and cohesive

elements to have the same mass density ρ . The speed of sound in the bulk and cohesive elements is then given by

$$c_b = \sqrt{\frac{E}{\rho}}, \quad c_c = \sqrt{\frac{K}{\rho}}. \quad (9)$$

Assuming small deformation, the effective speed of sound for the periodic system c_{eff} is defined as the periodic cell length in the reference configuration h divided by the total time t that it takes a stress wave to pass through both the solid and cohesive element. The result for the square periodic cell in Fig. 2 is

$$c_{\text{eff}} \equiv \frac{h}{t} = \frac{h}{\frac{h}{c_b} + \frac{d}{c_c}} = \frac{h}{\frac{h}{\sqrt{\frac{E}{\rho}}} + \frac{d}{\sqrt{\frac{K}{\rho}}}} = \frac{c_b}{1 + \frac{d}{h} \sqrt{\frac{E}{K}}}. \quad (10)$$

Similar to the effective stiffness in Eq. (5), the effect of cohesive element compliance is eliminated for $d \rightarrow 0$ or $K \rightarrow \infty$.

For a model meshed with triangular elements (see [supplementary material for details](#)), the c_{eff}/c_b ratio is

$$\frac{c_{\text{eff}}}{c_b} = \frac{1}{\frac{d}{h} \left[\frac{2\sqrt{B_b + \frac{4}{3}\mu_b}}{\sqrt{B_c + \frac{4}{3}\mu_c} + \sqrt{\mu_c}} + \sqrt{\frac{B_b + \frac{4}{3}\mu_b}{B_c + \frac{4}{3}\mu_c}} \right] + 1}, \quad (11)$$

where B_b , B_c , μ_b , μ_c are the bulk modulus and shear modulus of bulk elements and cohesive elements, respectively. For the special case where $B_b = E/3$, $B_c = K/3$, $\mu_b = E/2$, and $\mu_c = K/2$ (corresponding to a Poisson's ratio of zero) used in the numerical examples below, the ratio is

$$\frac{c_{\text{eff}}}{c_b} = \frac{1}{1 + \frac{d}{h} \sqrt{\frac{E}{K}} \left(\frac{3\sqrt{2} + 1}{\sqrt{2} + 1} \right)}. \quad (12)$$

2.4. Rescaling cohesive element properties for mesh independence

It is clear from the derivation in the previous section that adding cohesive elements changes the effective stiffness and effective speed of sound in the medium. Fig. 3(a) shows the dependence of E_{eff}/E and c_{eff}/c_b on K/E . Fig. 3(b) shows the dependence of E_{eff}/E and c_{eff}/c_b on the normalized cohesive element thickness d/h (both curves overlap).

Our objective is to determine the cohesive element properties K and d in terms of the Young's modulus E and the element size h , so that as the mesh is refined E_{eff} and c_{eff} remain constant and as close as possible to E and c_b . From Fig. 3(a) it is clear that changing K has different effects on E_{eff}/E and c_{eff}/c_b , whereas Fig. 3(b) shows that changing d has the same effect on both properties. Given this, in order to ensure mesh independence for a uniform mesh (i.e. no changes to the effective properties as the mesh size h is changed), it is recommended to rescale the cohesive element thickness so that d/h is kept constant. A value of $d/h = 0.05$ ensures that the effective properties are constant and within 95% of the bulk values for a quadrilateral mesh (91% for a triangular mesh as shown below).

3. Numerical results for the cohesive element rescaling method

To investigate the cohesive element rescaling method proposed in Section 2.4, we investigate the effects of mesh size and cohesive element parameters for four sample two-dimensional problems: (1) uniaxial stretching; (2) beam bending; (3) bar impact; and (4) crack propagation in Kalthoff plate impact. Simulations are performed using the Abaqus FEM code version 6.12 [11]. Cohesive

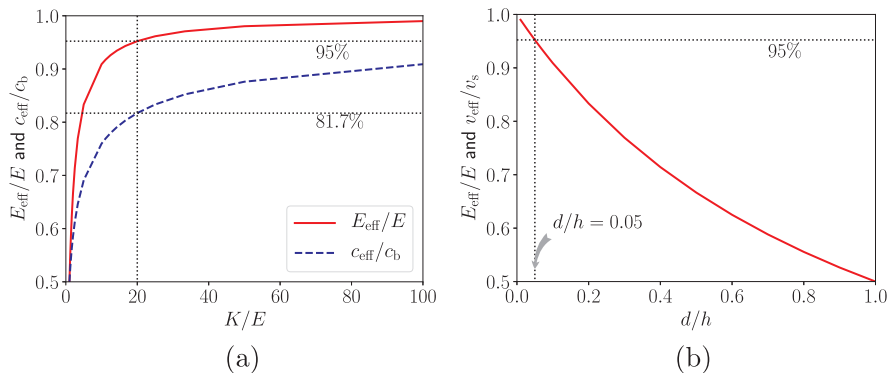


Fig. 3. Dependence of effective properties on cohesive element properties. (a) The solid curve is the relation between E_{eff}/E and K/E , the dashed curve is the relation between c_{eff}/c_b and K/E . (b) The relation between E_{eff}/E and c_{eff}/c_b and the normalized cohesive element thickness d/h (both curves overlap).

elements are inserted between all bulk elements using the RiCoh perl script [21].

3.1. Uniaxial stretching problem

Our first example is uniaxial stretching of a block as shown in Fig. 4(a). A 30 mm by 30 mm square block is meshed by triangular plane strain elements with different uniform mesh sizes (3 mm, 2 mm, 1 mm, 0.5 mm) as well as a nonuniform mesh with element size linearly grading from 3 mm to 1 mm. The block is composed of a material with Young's modulus $E = 3$ GPa and a Poisson's ratio $\nu = 0$ to eliminate the Poisson effect for simplicity. The cohesive element relation is the bilinear stress-separation law in Eq. (1) with $K = 3$ GPa and a corresponding relation for shearing with $\mu = 1.5$ GPa. The maximum stress (normal and shear) and fracture energy are taken to be a very large value so that the cohesive elements never fail. The bottom edge is fixed by rollers. An upward displacement of 0.1 mm is applied to the top edge, corresponding to an applied strain of $\epsilon_{\text{apply}} = 0.033$. The effective Young's modulus E_{eff} of the block is calculated from the FEM solution by dividing the stress in an element (all elements have the same stress) by the applied strain ϵ_{apply} .

Fig. 5 shows the dependence of E_{eff}/E on the normalized cohesive element thickness d/h for a mesh size of $h = 3$ mm. The numerical results obtained from the FEM simulations (triangles) are nearly identical to the analytical results (solid line) from Eq. (8). The results show that the smaller d/h , the closer E_{eff} is to E . For static calculations, in principal, d can be taken as small as desired. However, cohesive elements are typically used with explicit dynamics in which the time step is tied to smallest length scale which includes d . Therefore a compromise is required between accuracy in the effective properties and computational time. From Fig. 5 we see that taking $d/h = 0.05$ yields an effective stiffness within 10% of the bulk value, which is reasonably accurate and computationally tractable.

By rescaling the cohesive zone size d so that d/h is a constant for any mesh size, it is possible to obtain mesh independent results. Fig. 6(a) shows the effective stiffness as a function of mesh size for the four different mesh sizes shown in Fig. 6(b). The blue¹ curve (stars) shows that a constant effective stiffness is maintained when d/h is kept constant (i.e. the thickness is set to $d = 0.05h$). In contrast, the red curve (triangles) shows that keeping d constant leads to significant mesh dependence in the effective stiffness.

For a nonuniform mesh (e.g. the model shown in Fig. 7(b)), the cohesive element thickness d could be set to a different value for each element in order to obtain the constant $d/h = 0.05$ ratio. This is ideal as it would give a uniform stiffness throughout the model. However, this is complicated to do in Abaqus because of the way cohesive elements are defined in this code. Instead we set the cohesive element thickness of all elements to the same value d defined by $d/h_{\min} = 0.05$, where $h_{\min}$ is the smallest element size. This ensures that all elements have a stiffness E_{elem} that is least 90% of E . Fig. 7(a) shows the element stiffness measured in the simulation (normalized by E) as a function of element size. The smallest element has $E_{\text{elem}} = 0.9E$ and the larger the element, the closer E_{eff} is to E . The effective stiffness of the entire structure is 95.5%.

3.2. Beam bending problem

Our second example is bending of a cantilever beam as shown in Fig. 4(b). We estimate the effective Young's modulus E_{eff} from the beam end deflection relation,

$$E_{\text{eff}} = -PL^3/(3\delta I), \quad (13)$$

where P is the applied force, L is the length of beam, δ is deflection at the end of the beam, and I is the moment of inertia of the beam cross section. The beam is of length $L = 10$ mm and height $H = 1$ mm and out-of-plane width $B = 1$ mm and is meshed with triangular plane strain elements with a mesh size of 0.1 mm. Cohesive elements with different thicknesses (1 mm, 0.1 mm, 0.01 mm, and 0.001 mm) are inserted between every bulk element. The material properties of the bulk and cohesive elements are the same as those of the uniaxial stretching model in Section 3.1. The beam is fully constrained at the left end and is subject to a vertical concentrated force of 0.01 N downward at the right end.

Fig. 8(a) shows the beam end deflections obtained from simulations for different normalized cohesive element thicknesses. The horizontal dashed line is the result for a model with only bulk elements (no cohesive elements). Fig. 8(b) shows the dependence of the normalized effective stiffness calculated using Eq. (13) on the normalized cohesive element thickness d/h for a mesh of size $h = 0.1$ mm. The numerical results obtained from FEM simulations (triangles) are nearly identical to the analytical results (solid line) calculated from Eq. (8). As before, Fig. 8(b) shows that taking $d/h = 0.05$ yields an effective stiffness within 10% of the bulk value.

Similar to the uniaxial stretching model, we rescale the cohesive element thickness d keeping d/h constant for any mesh size. Fig. 9 shows the effective stiffness as a function of mesh size for three meshes (0.02 mm, 0.05 mm and 0.1 mm). The blue curve (solid line) shows the results for the scaled case (d/h kept constant). For comparison, the red curve (triangles) shows the results when keep d constant (so that d/h changes). As expected the end deflection, and hence the effective stiffness, is independent of mesh size when the cohesive elements are appropriately rescaled. (The small variation for $h = 0.1$ mm is due to the periodicity of the analytical model which includes an extra cohesive element relative to the finite model used in the simulation).

¹ For interpretation of color in the figures, the reader is referred to the web version of this article.

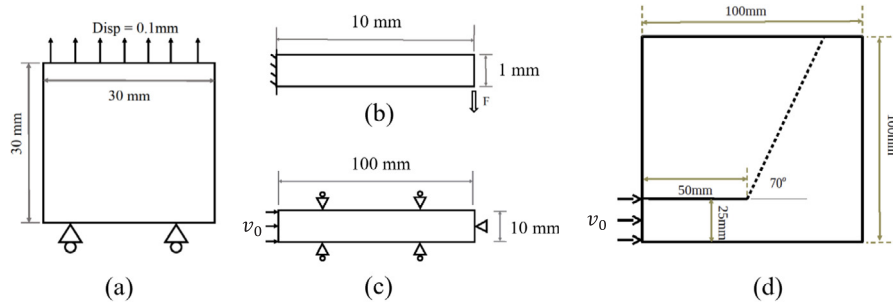


Fig. 4. Geometries of the tested models. (a) Uniaxial stretching. (b) Beam bending. (c) Bar impact. (d) Kalthoff plate impact experiment (the model represents only half of the full structure due to symmetry). The expected crack direction is shown as a dashed line.

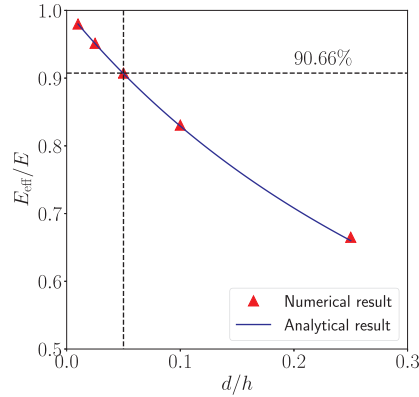


Fig. 5. Uniaxial stretching results for a mesh size of $h = 3$ mm. The graph shows the numerical and analytical results for E_{eff}/E at different d/h ratios. The analytical results are obtained from Eq. (8).

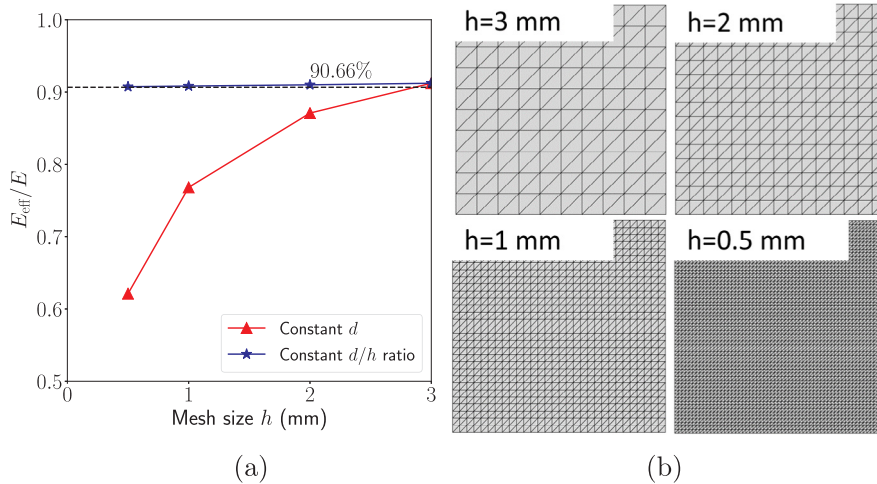


Fig. 6. Effective stiffness in uniaxial stretching for a uniform mesh. (a) The normalized effective stiffness as a function of mesh size. (b) The four meshes studied in (a).

3.3. Impact bar problem

The next example explores the effect of cohesive elements on a dynamical property, namely the effective longitudinal speed of sound c_{eff} in a 1D bar impact problem as shown in Fig. 4(c). A 100 mm by 10 mm bar is meshed by triangular plane strain elements with different mesh sizes (5 mm, 2.5 mm, 2 mm, 1 mm). The right end of the bar is pinned. The top and bottom edges are constrained by rollers to prevent buckling in the vertical direction. An initial velocity $v_0 = 0.01$ m/s is applied to the left end of the bar. Both bulk and cohesive elements have a density $\rho = 10^3$ kg/m³. Other material properties are the same as in the previous examples.

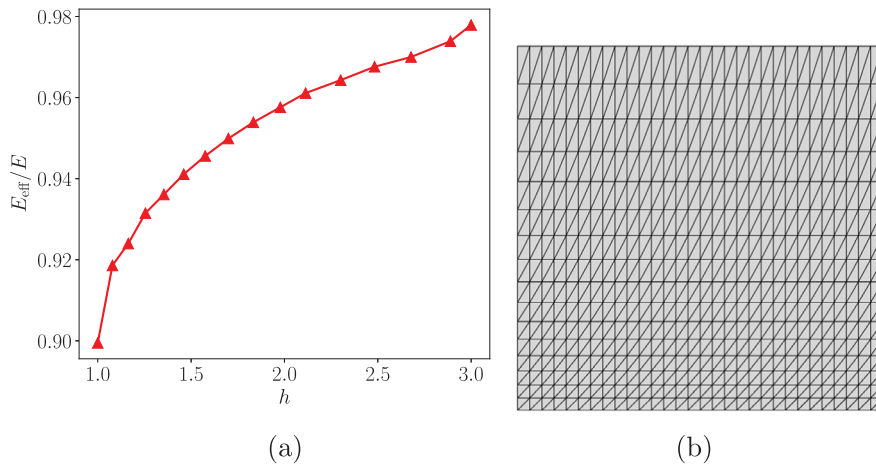


Fig. 7. Uniaxial stretching results for a nonuniform mesh. (a) Normalized element stiffness measured in the simulation as a function of element size. Each point is an average over the stiffness of all elements of that size. (b) The nonuniform mesh with elements ranging from $h = 1$ mm on the bottom to $h = 3$ mm on the top studied in (a).

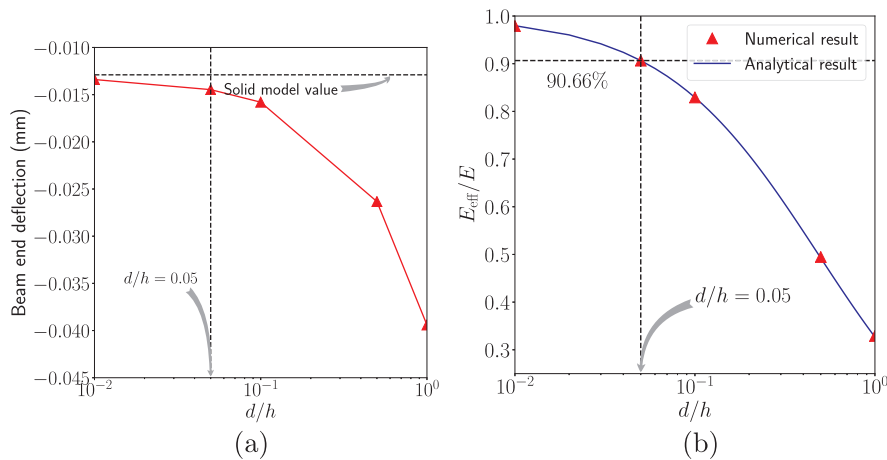


Fig. 8. (a) The relation between the deflection at the end of the beam and the d/h ratio. The horizontal dashed line is deflection for a bulk model without cohesive elements. (b) The relation between E_{eff}/E and d/h . Numerical and analytic results overlap.

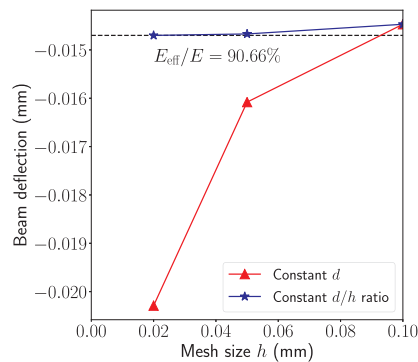


Fig. 9. Numerical results of beam end deflection with/without mesh rescaling method. The horizontal dashed line is -0.0147 mm, which is the beam end deflection corresponding to $E_{\text{eff}}/E = 90.66\%$.

The objective is to study how the effective speed of sound changes with the introduction of cohesive elements. First, we assess the speed of sound for a pure bulk specimen (i.e. without cohesive elements). Fig. 10(a) shows the velocity as a function of time at the midpoint of the bar (halfway along its length). A compressive wave is initiated at the left end and reaches the midpoint after about

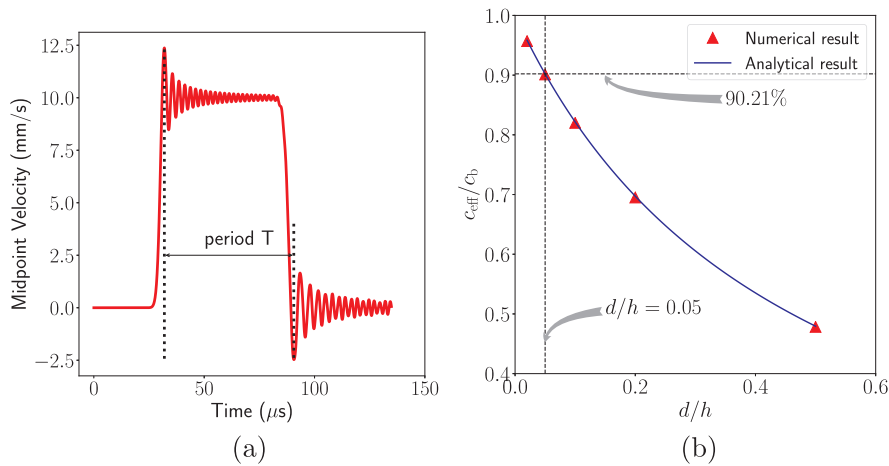


Fig. 10. (a) Velocity as a function of time at the midpoint of a 1D bar model (without cohesive elements) used to compute the speed of sound (see text). (b) The effect of cohesive elements on the speed of sound. Numerical and analytical results for c_{eff}/c_b as a function of the d/h ratio. Analytical results are obtained from Eq. (12).

30 μs . The wave then continues to the right end and reflects, reaching the midpoint again as a tensile wave after about 90 μs . The period $T = 57.75 \mu\text{s}$ between the first (positive) and second (negative) peaks reaching the midpoint is the time that it takes the stress wave to traverse the length of the bar ($L = 100 \text{ mm}$). The speed of sound is thus $c_b = L/T = 1731.6 \text{ m/s}$.

Next we introduce cohesive elements of thickness d between all elements and measure the effective speed of sound c_{eff} in this heterogeneous medium using the same approach described above for the pure bulk case. Fig. 10(b) shows the normalized effective speed of sound (c_{eff}/c_b) as a function of the normalized cohesive element thickness d/h for a mesh size of $h = 1 \text{ mm}$. The numerical results (triangles) are in excellent agreement with the analytical curve obtained from Eq. (12). Fig. 10(b) shows that taking $d/h = 0.05$ yields an effective speed of sound within 10% of the bulk value.

Fig. 11(a) shows the normalized effective speed of sound c_{eff}/c_b for the four mesh sizes shown in Fig. 11(b). The blue curve (stars) shows that a constant effective speed of sound is maintained provided that the ratio d/h is kept constant (i.e. the thickness is set to $d = 0.05h$). In contrast, the red curve (triangles) shows that keeping d constant leads to significant mesh dependence in the effective speed of sound. Thus the rescaling method provides mesh independent results for effective (average) dynamical properties.

3.4. Kalthoff plate impact problem

As a final example, we simulate a Kalthoff plate-impact experiment in which a plate containing a pair of edge cracks symmetrically arranged about the centerline is impacted by a rigid flat-ended projectile with velocity parallel to the centerline (Fig. 4(d) shows half the plate model with the projectile velocity uniformly distributed as an initial condition). This method was introduced by Kalthoff [22] and Kalthoff and Winkler [23] who used it to study crack propagation in C-300 maraging steel. They found crack extension at an angle of 70° to the centerline. Similar results were obtained numerically by Batra et al. [24]. The Kalthoff plate impact

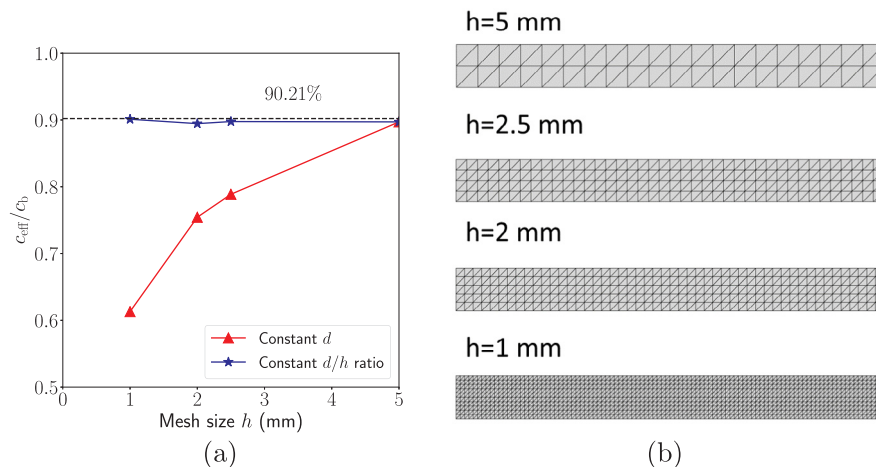


Fig. 11. (a) Effective longitudinal speed of sound as a function of mesh size in a 1D bar impact problem. (b) The four meshes studied in (a).

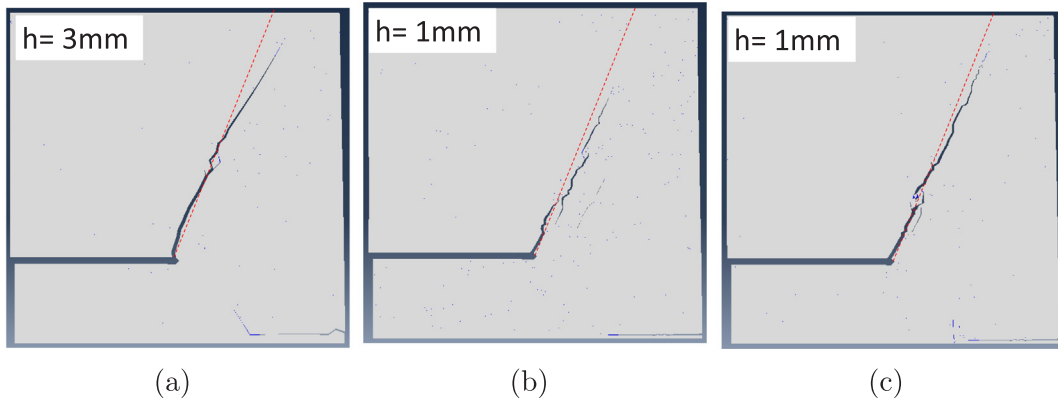


Fig. 12. Crack propagation for a model with two mesh sizes. (a) Mesh size $h = 3\text{ mm}$, cohesive element thickness $d = 0.15\text{ mm}$, $d/h = 0.05$. (b) $h = 1\text{ mm}$, $d = 0.15\text{ mm}$, $d/h = 0.15$. (c) $h = 1\text{ mm}$, $d = 0.05\text{ mm}$, $d/h = 0.05$.

problem is a popular benchmark problem for method development and has been used to evaluate cohesive element rescaling [19], XFEM [25] and phase field approaches [26].

In our problem, the plate is a rectangle of length 100 mm and height 200 mm, and the distance between the cracks is 50 mm. Due to symmetry, only half the problem is modeled as shown in Fig. 4(d). The projectile impact velocity is $v_0 = 20\text{ m/s}$, which is modeled by setting the initial velocity of the left edge of the model between the cracks to this value. The plate is made of steel with Young's modulus $E = 190\text{ GPa}$, Poisson's ratio $\nu = 0.3$ and mass density $\rho = 8\text{ g/cm}^3$. The cohesive zone moduli in the normal and tangential directions are $K_n = 190\text{ GPa}$ and $K_s = 73\text{ GPa}$, respectively, the yield stress $\sigma_{\max} = 400\text{ MPa}$, and the critical energy release rates in mode I and II under plane strain conditions are $G_{Ic} = G_{IIc} = 22.145\text{ kJ/m}^2$. The simulations were performed using explicit dynamics with a fixed time step of 1 ns for a duration of 0.075 ms.

Fig. 12 shows the crack propagation results for different mesh sizes and cohesive element thicknesses (element edges are not shown to highlight the crack). The red dashed line is at 70° to the centerline. Fig. 12(a) shows the results for a mesh size of $h = 3\text{ mm}$ and cohesive element thickness $d = 0.15\text{ mm}$ ($d/h = 0.05$). The crack grows by separating the model along cohesive elements. The growth direction is therefore limited by the mesh resolution, but is roughly aligned the 70° direction observed in experiments. We note the presence of a horizontal crack in the lower right corner. This is caused by a stress intensity at the artificial corner introduced by the symmetry conditions and can be ignored.

Fig. 12(b) shows the results for a mesh of size $h = 1\text{ mm}$ and the same cohesive element thickness as in Fig. 12(a) ($d/h = 0.15$). The finer mesh allows the crack to follow the 70° direction more closely than in Fig. 12(a), however note that the length of the crack is shorter. This is because the higher d/h ratio makes the model more compliant and hence able to absorb more energy prior to fracture. This mesh dependence can be addressed by rescaling the cohesive element size. Fig. 12(c) shows the results for a mesh of size $h = 1\text{ mm}$ with the cohesive element rescaled to $d = 0.05\text{ mm}$ so that the d/h ratio is the same as in Fig. 12(a) ($d/h = 0.05$). The total crack length in Fig. 12(a) and (c) are about the same. This is shown in quantitatively in Fig. 13 where the crack length is plotted as a function of time for the three models in Fig. 12. We see that the coarse mesh and refined mesh with rescaled cohesive elements are in

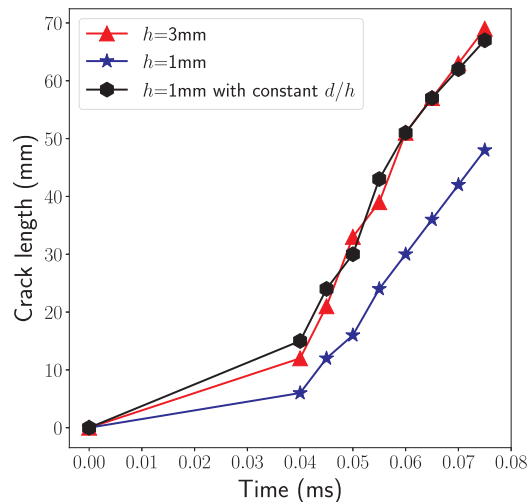


Fig. 13. Crack length as a function of time in a Kalthoff plate impact simulation.

close agreement, whereas refining the mesh without rescaling leads to a significant difference in crack propagation.

4. Conclusions

In order to simulate fracture within the FEM framework it is necessary for cracks to be able to form and grow. A popular approach is to introduce cohesive elements that fail under certain conditions between the other elements in the mesh allowing them to separate. This approach is physically reasonable and easy to implement, however, introduction of cohesive elements increases the compliance of the model leading to mesh dependence in the results. This is because cohesive elements are associated with an artificial thickness that effectively increases the size of the model. The change in compliance can be significant, especially when cohesive elements are inserted between all elements.

In this work, we show that mesh dependence can be minimized by suitably rescaling the cohesive element thickness d with mesh size, while keeping the cohesive element stiffness K constant. We derive analytical results for the effect of cohesive elements on the effective stiffness E_{eff} and effective speed of sound c_{eff} under 1D loading conditions. The results show that both E_{eff} and c_{eff} are decreased by the presence of cohesive elements. The larger K and the smaller d , the lesser the effect. Thus it is best to make the cohesive elements as thin as possible and/or as stiff as possible — however this leads to numerical difficulties. Based on our simulations, good performance was obtained by setting K equal to the Young's modulus of the bulk material and $d/h = 0.05$ where h is the element size. This results in values for E_{eff} and c_{eff} within 10% of bulk values for a triangular mesh (5% for a square mesh). Improved accuracy can be obtained by taking a smaller d/h ratio at an increased computational cost (especially for explicit dynamics simulations where the time step is tied to the smallest length scale which includes d). By maintaining d/h as a constant during mesh refinement (i.e. rescaling the cohesive element thickness with mesh size, $d = 0.05h$, mesh independent results are obtained for uniform meshes. For nonuniform meshes, rescaling by the minimum element size, $d = 0.05h_{\text{min}}$ minimizes mesh dependence effects.

The effect of cohesive elements and the proposed rescaling approach are tested in several example problems using the Abaqus finite element code: effective stiffness in uniaxial stretching and bending of a bar, effective speed of sound in a bar impact, and crack propagation speed and direction in Kalthoff plate impact. These problems are used to study the effect of cohesive element properties (K and d) and mesh refinement (h) on the results. For the 1D problems, the simulation results are in excellent agreement with the derived analytical expressions. In all cases, the proposed rescaling approach either eliminates (for uniform meshes) or significantly reduces mesh dependence for effective properties, such as average stiffness, speed of sound, and crack propagation speed and direction.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.engfracmech.2019.03.035>.

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